

Yale OpenHand model Q

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https://www.eng.yale.edu/grablab/openhand/model_q.html#about



<https://github.com/grablab/openhand-hardware/tree/master/model%20q>

1 BOM

1.1 Off the shelf parts

Part Name	Quantity	Description
Power Pro Spectra	1	Tendon
PMC-780 Urethane	1	Finger Joint Urethane
Vytaflex 30 Urethane	1	Finger Pad Urethane
Ø1/4", L2", 8-32 female standoff	4	Support
Ø1/4", L1" steel dowel pin	4	Joint/tendon routing pin
Ø1/4", L5/8" steel dowel pin	2	Tendon routing pin
Ø1/8", L5/8" steel dowel pin	8	Tendon routing pin
Ø1/8", L3/8" steel dowel pin	3	Tendon routing pin
8-32, L3/4" countersunk bolt	8	Support bolt
6-32, L0.15" heat-set insert	2	Insert for gear (NN)
6-32, L3/8" socket bolt	2	Fastener for gear (NN)
4-40, L0.135" heat-set insert	4 (anchor) + 4 (PP)	Insert for bolt anchors (AO anchor: any small nut)
4-40, L1/4" countersunk screw	4 (anchor) + 4 (PP)	Tendon/spring anchors (AO anchor: any small nut)
Any small nut	4	Tendon terminator (AO anchor: 4-40 screws and inserts)
M2, L5mm socket bolt	1	Horn fastener
Torsion spring, 0.340" OD, 0.028" wire diameter	2	Return spring (AO: extension springs)
Extension spring, 0.188" OD, L3/4", 0.016" wire diameter	2 + 2 (PP)	Return spring (AO: torsion spring at proximal). 2 required for PP.
Ø3/8", nylon pulley	1	Tendon-routing pulley
Dynamixel XM430-W350-R	4	Actuator

NN – Not Necessary for a well-functioning hand

AO – There exists an Alternative Option for the piece

PP – Required for pivot-pivot precision grasp fingers

PF – Required for pivot-flexure precision grasp fingers

1.1.1 Printed parts

Part Name	Quantity	Description
finger_ffff_q.stl	1	Power grasp fingers cutaway mold
See page 5 for PF finger options - OR - See page 7 for FF finger options	2	
top_outer.stl	1	Structure piece
pivot_base.stl	2	Base for precision fingers
finger_ffff_q.stl	1	Power finger mold
top_inner.stl	1	Base for power fingers
gear_main.stl	1	Main gear for power finger rotation
pulley_block_half.stl	2	Component of pulley block
mid_clamp.stl	1	Structure piece
gear_motor.stl	1	Gear to be attached to motor
horn.stl - OR - horn2.stl	3	Pulley horn
bottom_with_bridge.stl - OR - bottom_no_bridge.stl	1	Structure piece
bottom_reroute.stl	1	Tendon rerouting piece
bottom_brace_insert.stl - OR - bottom_brace_insertv2.stl - OR - bottom_brace_insertv3.stl	1	Arm mounting piece

2 Design notes

2.1 Under actuated tendon driven

Principle: Achieve useful in-hand manipulation with *fewer actuators* by combining two underactuated finger pairs with a central rotary degree of freedom.

- The creators describe Model Q as two tendon-driven, underactuated finger pairs “decoupled by an independent, central, rotating axis,” enabling finger-gaiting / regrasping while staying low-cost and compact.
- Design rationale: avoid fully anthropomorphic complexity; instead do a small set of robust, repeatable manipulation primitives (gaiting, twisting, pivoting).

Why: Underactuated fingers already give passive adaptability; adding a single rotary axis between pairs expands the reachable manipulation behaviours without adding per-finger ab/adduction joints (which would raise part count and control complexity).

2.2 Differentiated fingers for different job roles

Principle: Uses different joint technologies where they matter most.

- The hand uses a power-grasping pair with flexure-based proximal joints for surface conformity and robustness, and a precision-grasping pair with pin (revolute) proximal joints (preloaded with springs) to reduce out-of-plane compliance and improve pinch stability.
- The OpenHand finger library explicitly supports three joint-type combinations—Flexure-Flexure, Pivot-Pivot, Pivot-Flexure—with clear trade-offs (compliance/robustness vs twist-out vs precision).

Why: Flexures are great for tolerance to collisions and positioning error but can “twist out” in precision tasks; a proximal pin joint reduces that failure mode.

2.3 Design for Manufacturing: combining rigid prints + soft joints/pads

Principle: The creators print the rigid links, then cast the compliant joints.

- The Finger Guide lays out Hybrid Deposition Manufacturing (HDM) as: print molds → assemble → deposit urethanes/resins → disassemble.
- They use two-part urethanes: PMC-780 for flexures and Vytaflex for pads.
- The creators describe printing the finger bodies and mold shells, then pouring urethanes into joint/pad cavities; plus adding small anchoring features so urethane stays bonded to the printed body.

Why: This yields compliant, durable joints and high-friction pads without expensive multi-material printers or the need for additional spring components—while keeping geometry easily editable.

2.4 Design for Assembly: Choosing fasteners and joining methods that use simple tools

Principle: Assembly uses press-fits, standard dowel pins, heat-set inserts, and a small set of common fasteners.

- Model Q uses many steel dowel pins as joint pins and tendon routing pins and emphasises “press fit” installation.
- Heat-set inserts appear repeatedly (tendon anchors, gear attachment, etc.), but the assembly notes often note they’re not strictly necessary for a functioning hand—explicitly supporting simplified builds.
- Finger post-processing guidance is built around common workshop steps (reaming printed holes, drilling tendon ports, trimming sacrificial walls), reflecting “designing for real prints” rather than ideal CAD.

Why: Press-fit pins and inserts reduce alignment demands and keep the bill-of-process simple—no precision machining required.

2.5 Modularity and interchangeability

Principle: Tendon-driven fingers enable swapping whole finger sets and keeping the palm/actuation base consistent.

- The Finger Guide document explicitly says tendons make hands more modular and allow “entire sets of fingers [to be] interchanged on top of the same actuation base,” while staying compact.
- Model Q assembly lists multiple finger options and mold variants, and labels parts as AO (alternative option) / NN (not necessary) / PP/PF required for specific finger types, which provides the end user multiple choices.

Why: This supports research iteration: you can test different finger behaviours without redesigning the whole hand.

2.6 Robust tendon routing and wear management

Principle: Route tendons over steel dowel pins to avoid abrasion against printed plastic and use low-stretch line.

- The creators note Spectra tendon routed across steel dowel pins “to prevent abrasive wear and tear on the printed ABS surfaces.”
- The Finger Guide document recommends non-stretch, thin tendon (example: 100 lb Spectra) and highlights tendon tie-off as the main practical challenge, offering termination alternatives (nut/washer vs insert+bolt).

Why: Printed plastics wear quickly under sliding line; steel pins are cheap “sacrificial” wear surfaces.

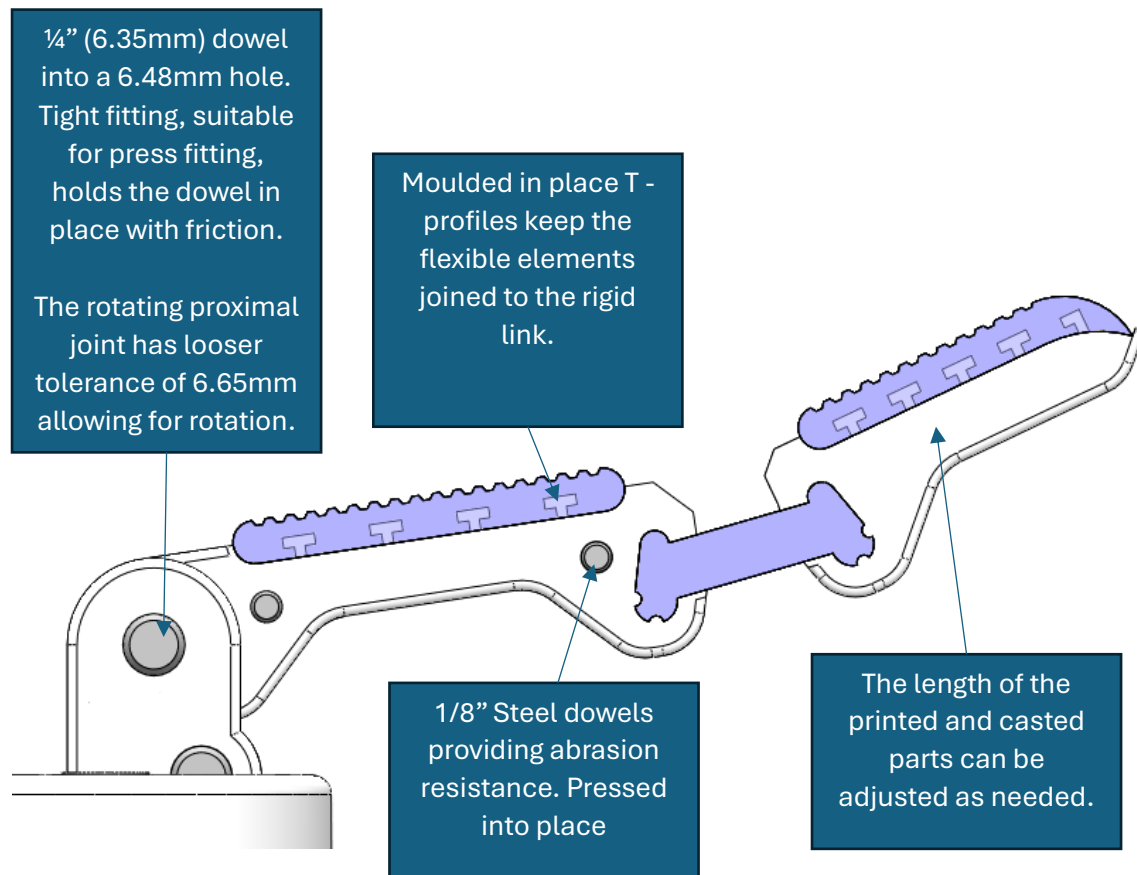
2.7 Tuneable passive compliance as a design parameter

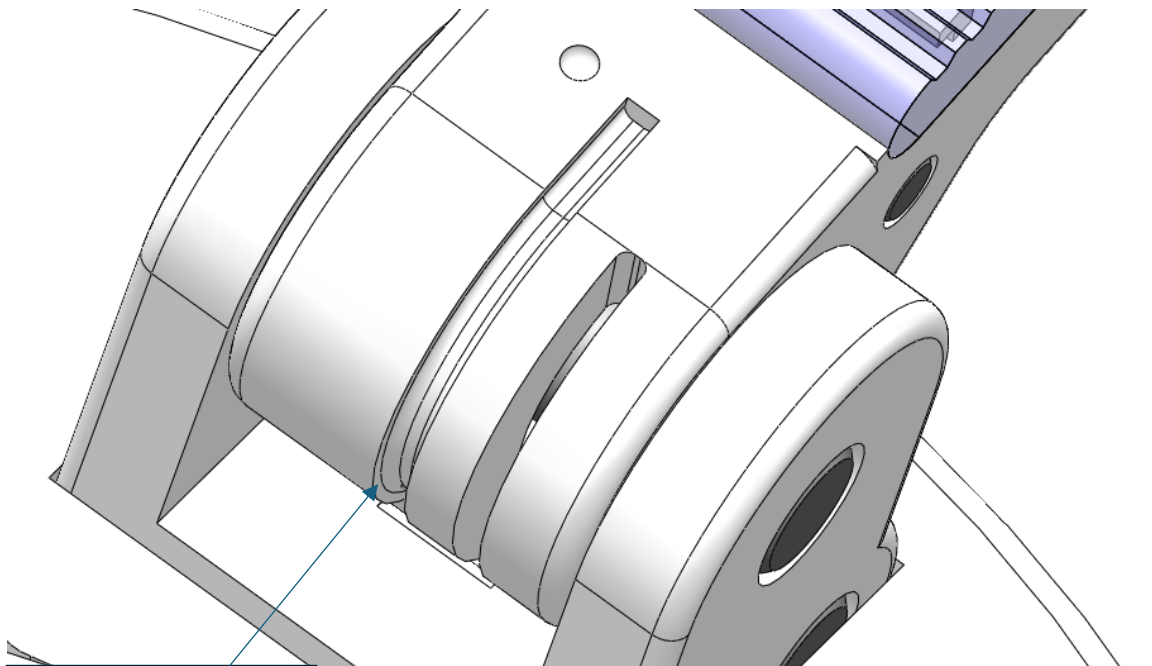
Principle: Use return springs (torsion/extension/bands) and targeted stiffness ratios to shape closing behaviour.

- Finger Guide document explains single-acting tendons require return action and discusses torsion vs extension spring trade-offs.
- The creators states design choices like:
 - proximal/distal link ratio ≈ 1.5 (human-like),
 - transmission ratio set ≈ 1 (empirical),
 - distal joint $\approx 2.5\times$ stiffer than proximal to encourage proximal contact first.

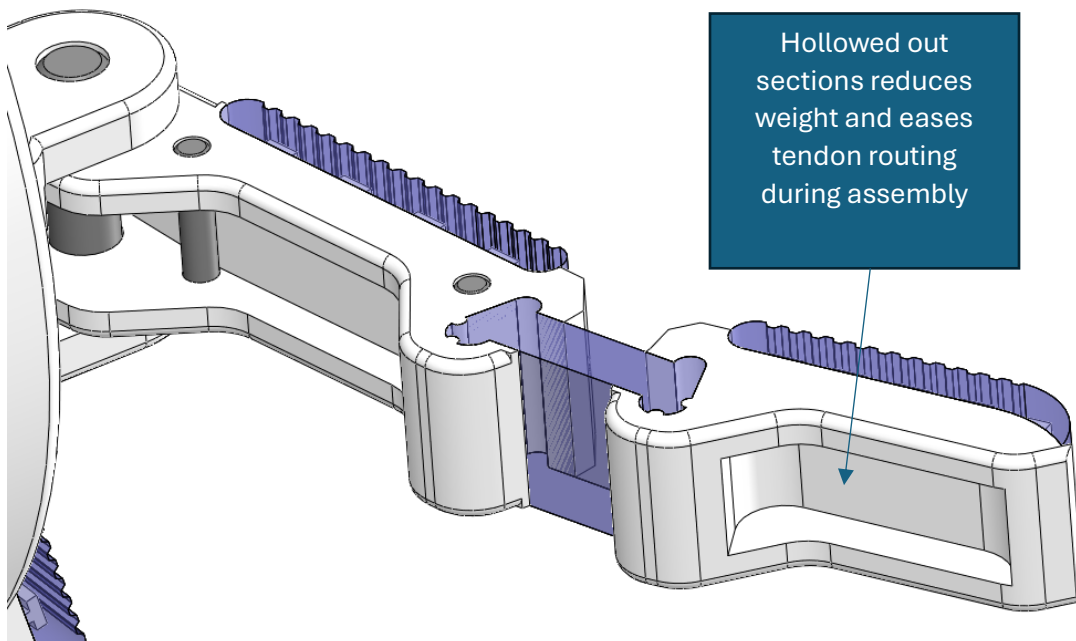
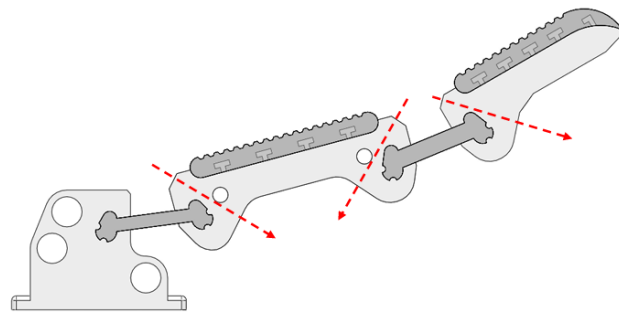
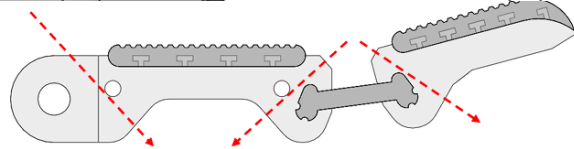
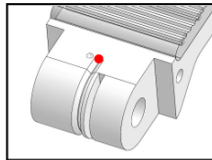
Why: In underactuated hands, behaviour emerges from compliance + contact—so stiffness is effectively part of the “control.”

3 CAD highlights

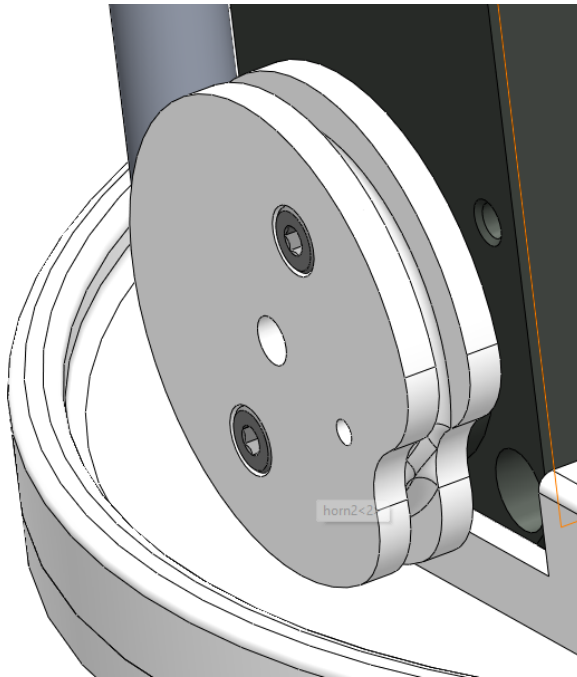




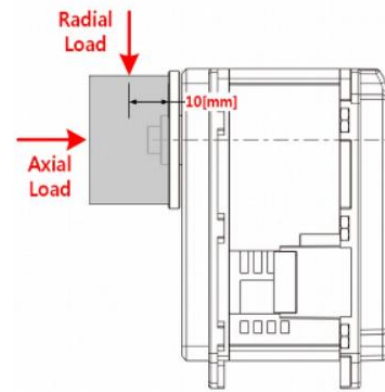
Tendon path not modelled into the CAD model. Post print steps include detail for drilling these out

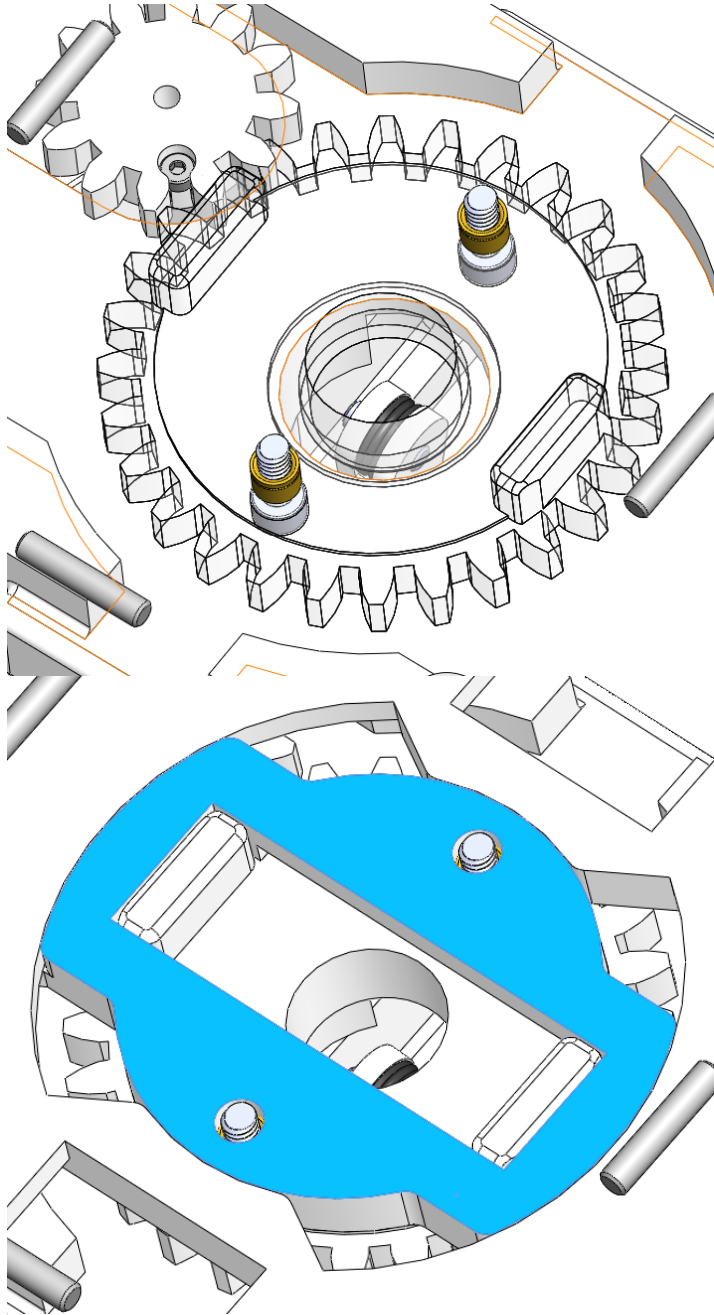


Hollowed out sections reduces weight and eases tendon routing during assembly



XM4330-w350-r
motors rated for 40N
radial load. No need
for double bracing of
the pulley.





No bearing
supporting the main
gear

Gear remains
concentric with this
'top_inner' part to
constrain
movement. 0.18mm
gap between sliding
surfaces provides
the room to rotate.